6D_1 is the Case where the two curled dimensions form a torus, T^2, the product of two circles. That is, $T^2 = S^1 x S^1$.

Note. T^2 is homogeneous but not isotropic. Thus, the solution is expected to have more free parameters than the 6D_2 case where the compact curled space is a 2-sphere, S^2.

Notation and Setup:

Starting Metric: $g_{00}=A[t, r]$, $g_{11}=-\frac{a[t]^2}{\mathcal{D}_5[r]}$, $g_{22}=-r^2\, a[t]^2$, $g_{33}=-r^2\, a[t]^2\, \text{Sin}[\theta]^2$, $g_{44}=-a[t]^2\, p_4^2$, $g_{55}=-a[t]^2\, p_5^2\, \mathcal{D}_5[r]$, a(t) not constant

Corresponding Line Element

$ds^2 = A[t, r]\, dt^2 - \frac{a[t]^2}{\mathcal{D}_5[r]}\, dr^2 - r^2\, a[t]^2\, d\theta^2 - r^2\, a[t]^2\, \text{Sin}[\theta]^2\, d\phi^2 - a[t]^2\, p_4^2\, d\psi_4^2 - a[t]^2\, p_5^2\, \mathcal{D}_5[r]\, d\psi_5^2$

Output Display Simplification - Parameters are sometimes suppressed
Prime (if any) means derivative wrt r.
Dot (if any) means derivative wrt t.
Examples: **A = A(t,r)**, $\mathbf{A' = \partial_r A(t,r)}$ and $\mathbf{\dot{A} = \partial_t A(t,r)}$

Lambda Vacuum Equations: The equations are $R_{\mu\nu} = \Lambda\, g_{\mu\nu}$. They are expressed as $\frac{R_{\mu\nu}}{g_{\mu\nu}} = \Lambda$ when $g_{\mu\nu} \neq 0$, and $R_{\mu\nu} = 0$ when $g_{\mu\nu} = 0$. The $g_{\mu\nu}$ matrix used is a diagonal matrix. Since off-diagonal $R_{\mu\nu}$ equations rarely appear, almost all the equations are expressed by dividing out $g_{\mu\nu}$.

GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameter a, (2) $\mathbb{C} = 1$, and (3) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation (which generates the a(t) formula).

FG GRW Equations: An exact solution to the Λ vacuum field equations that has (1) free parameters a and $\mathbb{C}$, and (2) parameter $\mathcal{A}$ such that $\mathcal{A}(t) = c^2$ is the Friedmann Continuity Equation.

a: GRW Equations

The line element above shows our initial assumptions. It explicitly shows

a(t) is a free parameter,

A(t,r) is a function of t and r,

$$B(r) = \frac{1}{\mathbb{D}_5(r)},$$

$$C(r) = \mathbb{D}_4(r) = 1,$$

and $\mathbb{D}_5(r)$ is a function of r.

Approach Overview:

- Like 5D, it is too difficult to solve for FG GRW first, so we begin with GRW.
  - ✓ By letting $C(r) = 1$, the equations are simplified and are solvable.
- The starting assumptions were derived in the appendix, which is a separate notebook.
  - ✓ Initially, parameters k, A, B, $\mathbb{D}_4$, and $\mathbb{D}_5$ were free parameters.
  - ✓ An intermediate solution was found that included two branches.
  - ✓ Branch 2 led to a dead end.
  - ✓ We investigate Branch 1 now.
  - ✓ In Branch 1, k, B, and $\mathbb{D}_4$ are no longer free and independent:
    - $B = \frac{1}{\mathbb{D}_5}$
    - $\mathbb{D}_4 = 1$
    - Curvature k = 0

In summary, we continue the Appendix investigation from the point where:

k = 0,

a(t) is not constant,

A(t,r) is a function of t and r,

$B(r) = \frac{1}{\mathbb{D}_5[r]}$, both strictly functions of r (or constants),

$C(r) = \mathbb{D}_4[r] = 1$.

The initial Lambda field equations are:

$$R_{00}: \quad \frac{5\,\ddot{a}}{a\,A} - \frac{5\,\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathbb{D}_5\,A'}{a^2\,A\,r} + \frac{\mathbb{D}_5\,(A')^2}{4\,a^2\,A^2} - \frac{A'\,\mathbb{D}_5'}{2\,a^2\,A} - \frac{\mathbb{D}_5\,A''}{2\,a^2\,A} = \Lambda$$

$$R_{01}: \quad -\frac{2\,\dot{a}\,A'}{a\,A} = 0$$

$$R_{11}: \quad \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} + \frac{\mathbb{D}_5\,(A')^2}{4\,a^2\,A^2} - \frac{\mathbb{D}_5'}{a^2\,r} - \frac{A'\,\mathbb{D}_5'}{4\,a^2\,A} - \frac{\mathbb{D}_5\,A''}{2\,a^2\,A} - \frac{\mathbb{D}_5''}{2\,a^2} = \Lambda$$

$$R_{22}: \quad \frac{1}{a^2\,r^2} + \frac{4\,\dot{a}^2}{a^2\,A} + \frac{\ddot{a}}{a\,A} - \frac{\dot{a}\,\dot{A}}{2\,a\,A^2} - \frac{\mathbb{D}_5}{a^2\,r^2} - \frac{\mathbb{D}_5\,A'}{2\,a^2\,A\,r} - \frac{\mathbb{D}_5'}{a^2\,r} = \Lambda$$

R_{33}: $\frac{1}{a^2 r^2} + \frac{4 \dot{a}^2}{a^2 A} + \frac{\ddot{a}}{a A} - \frac{\dot{a} \dot{A}}{2 a A^2} - \frac{\mathcal{D}_5}{a^2 r^2} - \frac{\mathcal{D}_5 A'}{2 a^2 A r} - \frac{\mathcal{D}_5'}{a^2 r} = \Lambda$

R_{44}: $\frac{4 \dot{a}^2}{a^2 A} + \frac{\ddot{a}}{a A} - \frac{\dot{a} \dot{A}}{2 a A^2} = \Lambda$

R_{55}: $\frac{4 \dot{a}^2}{a^2 A} + \frac{\ddot{a}}{a A} - \frac{\dot{a} \dot{A}}{2 a A^2} - \frac{\mathcal{D}_5'}{a^2 r} - \frac{A' \mathcal{D}_5'}{4 a^2 A} - \frac{\mathcal{D}_5''}{2 a^2} = \Lambda$

R_{33} is redundant, the same as R_{22}.

Since a(t) ≠ constant, from R_{01} we get that

A(t,r) = $\mathcal{A}$(t), a new unknown, strictly a function of t, which again tells us that A(t,r) cannot be a function of r.

R_{01} is eliminated, and the remaining equations become

R_{00}: $\frac{5 \ddot{a}}{a \mathcal{A}} - \frac{5 \dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$

R_{11}: $\frac{4 \dot{a}^2}{a^2 \mathcal{A}} + \frac{\ddot{a}}{a \mathcal{A}} - \frac{\dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{\mathcal{D}_5'}{a^2 r} - \frac{\mathcal{D}_5''}{2 a^2} = \Lambda$

R_{22}: $\frac{1}{a^2 r^2} + \frac{4 \dot{a}^2}{a^2 \mathcal{A}} + \frac{\ddot{a}}{a \mathcal{A}} - \frac{\dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{\mathcal{D}_5}{a^2 r^2} - \frac{\mathcal{D}_5'}{a^2 r} = \Lambda$

R_{44}: $\frac{4 \dot{a}^2}{a^2 \mathcal{A}} + \frac{\ddot{a}}{a \mathcal{A}} - \frac{\dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} = \Lambda$

R_{55}: $\frac{4 \dot{a}^2}{a^2 \mathcal{A}} + \frac{\ddot{a}}{a \mathcal{A}} - \frac{\dot{a} \dot{\mathcal{A}}}{2 a \mathcal{A}^2} - \frac{\mathcal{D}_5'}{a^2 r} - \frac{\mathcal{D}_5''}{2 a^2} = \Lambda$

R_{55} is now redundant, the same as R_{11}.

We can immediately find $\mathcal{A}$ from R_{00}:

$$\mathcal{A}(t) = \frac{5 \dot{a}[t]^2}{\Lambda a[t]^2}$$

where the constant of integration, k, has been set to zero (per assumptions).

R_{00} is eliminated, and the remaining equations become

R_{11}: $\Lambda - \frac{\mathcal{D}_5'}{a^2 r} - \frac{\mathcal{D}_5''}{2 a^2} = \Lambda$

R_{22}: $\frac{1}{a^2 r^2} + \Lambda - \frac{\mathcal{D}_5}{a^2 r^2} - \frac{\mathcal{D}_5'}{a^2 r} = \Lambda$

R_{44}: $\Lambda = \Lambda$

R_{44} is an identity, and we can solve R_{22} for $\mathcal{D}_5$:

$\mathcal{D}_5(r) = 1 + \frac{\chi}{r}$ where χ is a constant of integration.

This satisfies remaining equation

R_{11}: $\Lambda = \Lambda$

All of the equations are satisfied, curvature k = 0,

and the GRW solution is

$$A(t,r) = \mathcal{A}(t) = \frac{5\,\dot{a}[t]^2}{\Lambda\,a[t]^2} = c^2 \quad \text{where } \Lambda > 0$$

$$B(r) = \frac{1}{\mathcal{D}_5[r]},$$

$$C(r) = 1,$$

$$\mathcal{D}_4(r) = 1,$$

$$\mathcal{D}_5(r) = 1 + \frac{\chi}{r} \quad \text{where } \chi \text{ is a constant of integration,}$$

and line element

$$ds^2 = c^2\,dt^2 - \frac{a[t]^2}{1+\frac{\chi}{r}}\,dr^2 - r^2\,a[t]^2\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,d\phi^2 - a[t]^2\,p_4^2\,d\psi_4^2 - \left(1+\frac{\chi}{r}\right)a[t]^2\,p_5^2\,d\psi_5^2$$

Notes:

- Radii p_4 and p_5 remain completely free and independent of each other.
  - ✓ The two radii of the torus need not be the same size.
  - ✓ The aspect ratio $\frac{p_4}{p_5}$ is a genuine 'modulus' of the solution:
    - ○ It is a flat direction, undetermined by the vacuum-Λ field equations alone.
  - ✓ This is the same phenomenon as the radion/moduli-stabilization problem we see in Kaluza-Klein compactifications.

b: Solve $\mathcal{A}(t) = c^2$ to find a(t)

Solving $\frac{5\,\dot{a}[t]^2}{\Lambda\,a[t]^2} = c^2$ yields

$$a(t) = e^{\pm\frac{c\,t\,\sqrt{\Lambda}}{\sqrt{5}}}\,\alpha \quad \text{where } \alpha > 0 \text{ is a constant of integration and } \Lambda > 0.$$

c: Kaluza-Klein (KK) bubble-of-nothing

Case 6D_1 has two curled circles, parameterized by ψ_4 and ψ_5.

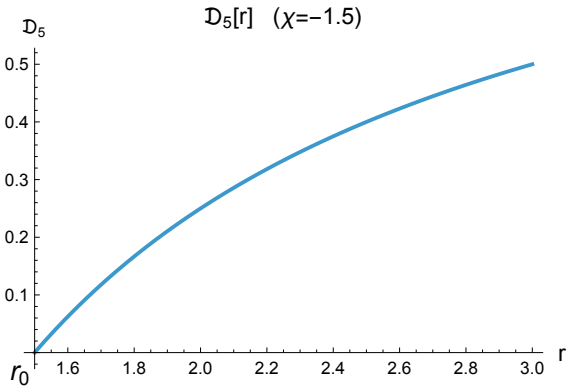
- The circles begin at ψ_μ=0. Each winds around the circumference of a circle of radius p_μ and arrives back at the starting point at ψ_μ=2π.
- The circles thus have period 2π.

The GRW solution $\mathcal{D}_5(r) = 1 + \frac{\chi}{r}$ ties the ψ_5 circle to r, even though they were initially unrelated. There is no such tie between the ψ_4 circle and r because $\mathcal{D}_4(r) = 1$.

Define $r_0 \equiv -\chi$.

In order for $\mathcal{D}_5(r) = \frac{1}{B[r]}$ to be positive, we must have $r > r_0$.

The sample $\mathcal{D}_5$ plot below (for χ=-1.5, r_0=1.5) helps visualize the domain of r.



This restriction on the domain of $\mathcal{D}_5$ is the 6D_1 analog of RCPC.

$r_0 = -\chi$ is the 'pinch' point:

- $\mathcal{D}_5(r_0) = 0$.
- For $\chi < 0$
  - ✓ $r_0 > 0$
  - ✓ the 6D_1 RCPC domain of r is restricted to $r > r_0$.
  - ✓ As $r \to r_0$ from above, $\mathcal{D}_5 \to 0$.
- For $\chi > 0$, $\mathcal{D}_5 > 0$ and so there is no domain restriction.

Analogous to the original KK bubble, the radial line element for this solution is

$$ds^2 = \frac{1}{\mathcal{D}_5[r]}\, dr^2 + \mathcal{D}_5[r]\, p_5^2\, d\psi_5^2$$

This is a good time to introduce some vocabulary.

The full 6D spacetime contains 6 parameters, $(t,r,\theta,\phi,\psi_4,\psi_5)$ and has an associated line element. The portion of the line element involving 3-space, (r,θ,ϕ) is called the **base line element**. Any other portion of the line element, such as the (r,ψ_5) line element, above, is referred to as a **transverse line element**.

In flat space, $ds^2 = dr^2$; i.e., the radial distance r equals the proper distance s. In curved space, they are not the same. Hence, to check whether the ψ_5 circle shrinks smoothly down to zero, we have to use the proper distance, s.

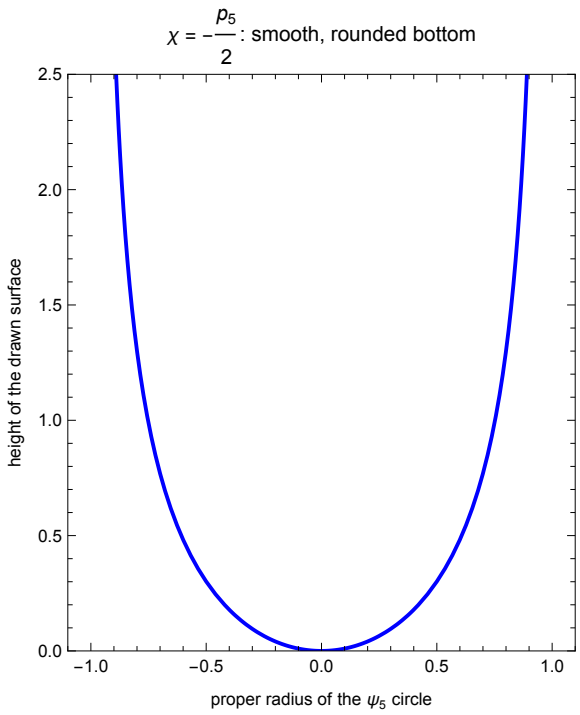
$\mathcal{D}_5[r]\ p_5^2$ is the square of the proper radius. So, our plan is to express $\sqrt{\mathcal{D}_5[r]}\,p_5$, the proper radius, in terms of s, and then to examine the proper circumference, $2\pi\sqrt{\mathcal{D}_5[r]}\,p_5$.

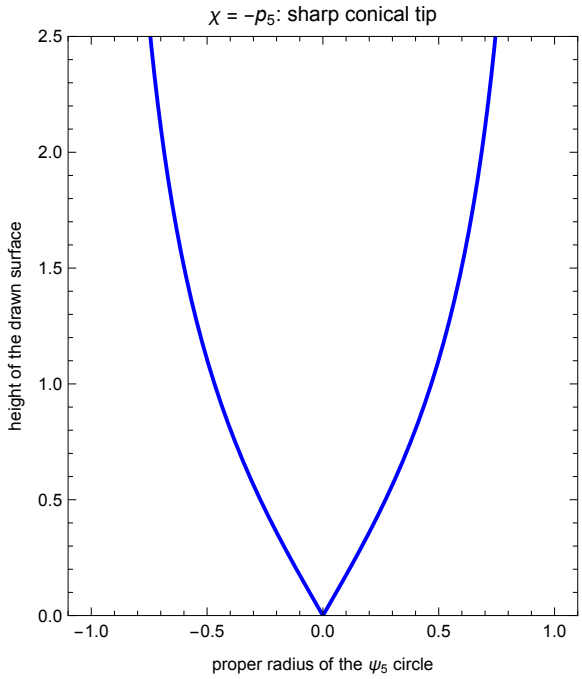
As $r\to r_0$, the ψ_5-circle circumference $2\pi\sqrt{\mathcal{D}_5[r]}\,p_5$ shrinks down to zero. This is the bubble: the curled ψ_5-direction pinches off completely at $r=r_0$. As r moves away from r_0, spacetime continues outward.

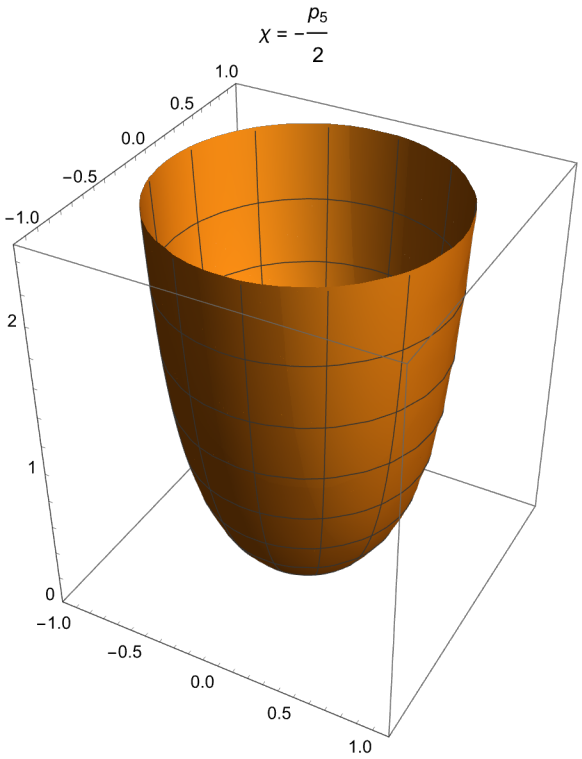
Notice that as $r\to r_0$, $\mathcal{D}_5\to 0$, the $\frac{dr^2}{\mathcal{D}_5}$ term in ds^2 gets large and the $\mathcal{D}_5 p_5^2 d\psi_5^2$ term gets small, so that $ds^2 \approx \frac{dr^2}{\mathcal{D}_5[r]} = \frac{dr^2}{1+\frac{\chi}{r}}$. This is the metric we use to determine whether the pinch has a sharp point or is smooth.

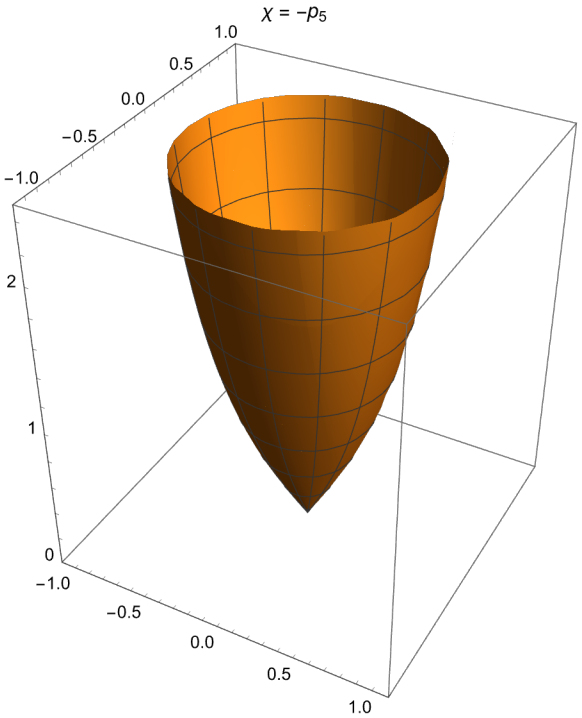
The plots below help visualize the pinch. In the plots, $\sqrt{\mathcal{D}_5}\,p_5$ are the proper radii of the horizontal slices for various r-values. The upper plots show 2D cross-sections of the height of the surface vs proper radii. The lower plots revolve the upper plots to form surfaces of revolution in ordinary 3-space.

With $\chi=-\frac{p_5}{2}$ the bottom is rounded (smooth); with $\chi=-p_5$ it comes to a sharp point (a cone). The claim below show that $\chi=-\frac{p_5}{2}$ is the only value that gives a smooth bottom. (Note: the plot uses $p_5=1$.)









Claim: smoothness of the pinch places a constraint on χ: $\chi = -\frac{p_5}{2}$

$$\Rightarrow \quad \mathcal{D}_5[r] = 1 + \frac{\chi}{r} = 1 - \frac{p_5}{2\,r}.$$

Proof of claim:

Slope of the tangent line of $\mathcal{D}_5$ at r_0 is

$$m = \mathcal{D}_5'(r_0) = -\frac{\chi}{r^2}\Big|_{r_0} = -\frac{\chi}{(-\chi)^2} = -\frac{1}{\chi}.$$

For r near r_0, the tangent line is a 1st order approximation to $\mathcal{D}_5$.

Set $r \equiv r_0 + \delta_r$ for r near to r_0. Since $\mathcal{D}_5[r_0]=0$, using $y-y_0=m(x-x_0)$ we get,

$$\mathcal{D}_5[r] = \mathcal{D}_5[r]-\mathcal{D}_5[r_0] \approx m(r-r_0) = m\,\delta_r.$$

$$\Rightarrow ds \approx \frac{dr}{\sqrt{\mathcal{D}_5[r]}} \approx \frac{dr}{\sqrt{m\,\delta_r}}$$

$$\Rightarrow \quad s \approx \int_{r_0}^{r} \frac{1}{\sqrt{m\,\delta_\rho}}\,d\rho = \int_{r_0}^{r} \frac{1}{\sqrt{m\,(\rho - r_0)}}\,d\rho = \frac{2\sqrt{m\,(r - r_0)}}{m} = 2\sqrt{\frac{\delta_r}{m}}$$

$$\Rightarrow \delta_r \approx \frac{m\,s^2}{4}$$

$$\Rightarrow \mathcal{D}_5(r) \approx m\,\delta_r = \frac{m^2\,s^2}{4}$$

We have expressed $\mathcal{D}_5$ in terms of proper distance, s.

We use this to determine the ψ_5 circle's proper physical circumference.

Recall from above that $\sqrt{1+\frac{\chi}{r}}\,p_5$ is the proper physical radius for the ψ_5 circle.

In terms of s (rather than r),

$$\text{Proper radius} = \sqrt{1+\frac{\chi}{r}}\,p_5 \approx \frac{1}{2}\,m\,s\,p_5$$

Since ψ_5 sweeps its full range of 2π, the actual physical circumference is

$$\text{Proper circumference} = 2\pi\ \text{Proper radius} = m\,\pi\,s\,p_5$$

If the circumference doesn't equal 2πs exactly, the point isn't smooth:

- Less than 2πs is a sharp, cone-like point (a deficit angle)
- More than 2πs is a saddle-like excess.

Smoothness requires the circumference to equal exactly 2πs.

We now apply the smoothness constraint:

$$m\,\pi\,s\,p_5 = 2\,\pi\,s$$

$$\Rightarrow \quad \chi = -\frac{1}{m} = -\frac{p_5}{2}$$

This concludes proof of the claim.

Notes

(1) To claim smoothness, the full ds^2 line element involving t, r, Θ, ϕ, ψ_4, and ψ_5 must be considered. The transverse line element used to prove the claim involves just r and ψ_5. However since there are no off-diagonal metric terms $g_{\mu\nu}$, there is no mixing of different directions. Consequently, the entire 6D spacetime is smooth if the transverse space is.

(2) The $\frac{\chi}{r}$ term is replaced in the $\mathcal{D}_5(r)$ pattern by $-\frac{p_5}{2\,r}$ because χ is no longer a free constant.

(3) p_5, by contrast, remains a free constant, as does p_4.

d: Free-gauge GRW

The table below shows the GRW solution, obtained in Section a, and the FG GRW solution we are about to obtain. It also provides comparison to prior cases.

		4D	5D	6D_1
GRW	Constraints	None	None	$\Lambda>0,\ k=0$
	KK-bubble Con-straints (optional)	n/a	$\chi=r_0(-1+kr_0^2)$ where $r_0=\frac{\mp1+\sqrt{1+3\,k\,p_4^2}}{3\,k\,p_4}$ and $k\geq-\frac{1}{3\,p_4^2}$	$\chi=-\frac{p_5}{2}$
	$\mathcal{A}$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$	$\mathcal{A}(t)=c^2$
	B	$\frac{1}{1-kr^2}$	$\frac{1}{\mathcal{D}_4}$	$\frac{1}{\mathcal{D}_5}$
	C	1	1	1
	$\mathcal{D}_4$		$1-k\,r^2+\frac{\chi}{r}$	1
	$\mathcal{D}_5$			$1-k\,r^2+\frac{\chi}{r}$
FG GRW	$\mathcal{A}$	$\frac{3\,\dot{a}^2}{-3\,k+a^2\,\Lambda}$	$\frac{4\,\dot{a}^2}{-4\,k+a^2\,\Lambda}$	$\frac{5\,\dot{a}^2}{-5\,k+a^2\,\Lambda}$
	B	$\frac{(2\,C+r\,C')^2}{4\,C\,(1-k\,r^2\,C)}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_4}$	$\frac{(2\,C+r\,C')^2}{4\,C\mathcal{D}_5}$
	C	C (free)	C (free)	C (free)
	$\mathcal{D}_4$		$1-k\,r^2\,C+\frac{\chi}{r\sqrt{C}}$	1
	$\mathcal{D}_5$			$1-k\,r^2\,C+\frac{\chi}{r\sqrt{C}}$

Notes:

- $\mathcal{A}(t)=c^2$ is the Friedmann Continuity Equation
  - ✓ The solution to the equation is the $a(t)$ formula from Section b
- The two constraint rows display restrictions on parameter constants
  - ✓ Due to the field equations
  - ✓ Due to KK-bubble smoothness
- By contrast, domain restrictions on r (RCPC and its analogs) are not displayed as constraints
- FG GRW has the same constraints (both rows) as GRW

The 6D Free-gauge GRW results were obtained from GRW by using the 5D pattern:

$\mathcal{A}$: Replace c^2 by $\mathcal{A}$ from the 6D GRW equations

B: Replace $r\to r\sqrt{C}$ and multiply result by $\frac{(2\,C+r\,C')^2}{4\,C}$

$\mathbb{C}$: Replace 1 by free parameter $\mathbb{C}$

$\mathbb{D}_4$: Replace r by $r\sqrt{\mathbb{C}}$ (nothing to replace)

$\mathbb{D}_5$ is new, so there is no established pattern. What works is simply

$\mathbb{D}_5$: Replace r by $r\sqrt{\mathbb{C}}$

The line element is

$$ds^2 = \frac{5\,\dot{a}[t]^2}{\Lambda\,a[t]^2}\,dt^2 - \frac{a[t]^2\,(2\,\mathbb{C}[r] + r\,\mathbb{C}'[r])^2}{4\left(1 - \frac{p_5}{2\,r\,\sqrt{\mathbb{C}[r]}}\right)\mathbb{C}[r]}\,dr^2 - r^2\,a[t]^2\,\mathbb{C}[r]\,d\theta^2 - r^2\,a[t]^2\,\text{Sin}[\theta]^2\,\mathbb{C}[r]\,d\phi^2$$
$$- a[t]^2\,p_4^2\,d\psi_4^2$$
$$- a[t]^2\,p_5^2\left(1 - \frac{p_5}{2\,r\,\sqrt{\mathbb{C}[r]}}\right)d\psi_5^2$$

Confirming:

R_{00}: $\Lambda = \Lambda$

R_{01}: $0 = 0$

R_{02}: $0 = 0$

R_{03}: $0 = 0$

R_{04}: $0 = 0$

R_{05}: $0 = 0$

R_{11}: $\Lambda = \Lambda$

R_{12}: $0 = 0$

R_{13}: $0 = 0$

R_{14}: $0 = 0$

R_{15}: $0 = 0$

R_{22}: $\Lambda = \Lambda$

R_{23}: $0 = 0$

R_{24}: $0 = 0$

R_{25}: $0 = 0$

R_{33}: $\Lambda = \Lambda$

R_{34}: $0 = 0$

R_{35}: $0 = 0$

R_{44}: $\Lambda = \Lambda$

R_{45}: $0 = 0$

R_{55}: $\Lambda = \Lambda$

e: The Vacuum Friedmann Equations

Plugging $\mathcal{A}$, B, C, and D_4 (but not a) from the GRW solution into the field equations yields:

$$R_{00}: \quad \frac{5\,\ddot{a}}{a\,c^2} = \Lambda$$

$$R_{11}: \quad \frac{4\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$$

$$R_{22}: \quad \frac{4\,\dot{a}^2}{a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \Lambda$$

R_{22} is redundant, the same as R_{11}.

The Friedmann continuity and acceleration equations, respectively, are the following pair of equations generated by algebraic manipulation of R_{00} and R_{11}:

$$-\frac{1}{20}R_{00}+\frac{1}{4}R_{11}: \quad \frac{\dot{a}^2}{a^2\,c^2} = \frac{\Lambda}{5}$$

$$\frac{1}{6}R_{00}+\frac{1}{6}R_{11}: \quad \frac{2\,\dot{a}^2}{3\,a^2\,c^2} + \frac{\ddot{a}}{a\,c^2} = \frac{\Lambda}{3}$$

As pointed out earlier, $\mathcal{A}(t) = c^2$ is simply a rearrangement of the continuity equation. Since the a(t) formula in Part a is the result of solving $\mathcal{A}(t) = c^2$, the 4-branch a(t) formula is also the solution to the Friedmann equations.